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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT601/20BT601 – PRINCIPLES OF FERMENTER DESIGN AND CONTROL

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I**, **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) A stirred tank bioreactor is to be designed for vinegar production. Suggest the preferred material for bioreactor fabrication. Also give the geometry ratios for the bioreactor.	10	L*2	1	3
	b) List and explain the various sterilization methods adapted for bioreactor.	10	L2	2	3
2.	a) Explain the step wise operation of batch fermenter for microbial cultivation	10	L2	1	3
	b) What is the requirement of aseptic condition inside the bioreactor? Explain various aseptic measures to be considered during designing of bioreactor.	10	L3	2	3
3.	a) List and explain the different applications of valves used in bioreactor.	10	L2	1	3
	b) List and explain the various cleaning strategies adapted for fermenters.	10	L2	2	3
Unit – II					
4.	a) Derive an expression to determine the batch time required to reduce the substrate concentration from S_0 to S_f . Assume Michaelis-Menton kinetics.	10	L3	3	2
	b) A bioprocess in which oxygen requirement for aerobic cell culture is maintained by continuous sparging of sterile air. Derive an expression to determine the rate of mass transfer between liquid culture and gaseous oxygen.	10	L3	4	3
5.	a) A $5m^3$ fermenter is operated continuously with feed substrate concentration $20kg/m^3$. The microorganism cultivated in the reactor has the following characteristics, $\mu_{max} = 0.45/h$, $K_s = 0.8kg/m^3$, $Y_{xs} = 0.55kg/kg$.				
	i. What feed flow rate is required to achieve 90% substrate conversion?				
	ii. How does the biomass productivity at 90% substrate conversion compare with the maximum possible?	10	L3	3	3
	b) What are the difficulties associated with measuring oxygen transfer rates and k_{La} in large fermenters? Explain.	10	L2	4	3
6.	a) Determine the volumetric flowrate required to prevent cell washout in chemostat having Vm^3 .	10	L3	3	3
	b) List different types of spargers and its design consideration.	10	L3	4	3

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Unit – III

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| 7. a) | In detail explain the working of calomel electrode and polarographic electrode. | 10 | L2 | 5 | 2 |
| b) | A proportional (P) controller is used to control the liquid temperature in a fermenter. Obtain $C(s)/R(s)$ and find the expression for offset, when unit step change is given to the input. | 10 | L3 | 5 | 2 |
| 8. | Write a short note on (i) On/OFF Controller (ii) Feedback Controller (iii) Ratio Controller (iv) SCADA . | 20 | L2 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
